Auditable Measurement of Contested Legal Constructs with Large Language Models

Zhenling Chen

***Abstract*—Many legal systems rely on doctrinal constructs that are texturally open, fact-sensitive, and normatively contested. In such settings, the core methodological difficulty is not merely sparse labeling, but the absence of a single stable gold standard against which measurement can simply be benchmarked. This paper develops and validates a theory-driven, auditable framework for the continuous measurement of contested legal constructs. Rather than delegating the task to an end-stage judgment model, the framework uses large language models to extract fine-grained semantic indicators, organizes those indicators into explicit legal dimensions, and aggregates them through preset rules into a continuous index.**

As a stress-test case, the paper studies expansion risk in the Chinese offense commonly translated as "picking quarrels and provoking trouble." The paper makes three contributions. First, it provides an auditable legal-measurement design that translates a fuzzy doctrinal concern into a continuous and interpretable index. Second, it develops a layered validation strategy for weak-gold-standard legal settings, combining blind human audit, prompt-based direct AI comparison, corpus-scale deployment, and robustness checks into a convergent evidentiary structure. Third, it demonstrates the framework's empirical usefulness in a highly disputed criminal-law domain: on a 100-case blind-audit sample, the index shows a clear positive association with human judgment; under controlled prompt-based comparisons, direct end-stage AI judgment proves highly sensitive to whether cases are judged singly or in grouped reference sets; and on a 9000-case corpus, the framework preserves a selective upper tail, recovers structured expansion pathways, and responds to historically meaningful variation in legal context.

The paper does not claim to recover a final ground truth for a disputed doctrinal object. Its claim is narrower but stronger: contested legal constructs can be operationalized into auditable continuous measures that remain interpretable, debatable, and empirically testable without collapsing into opaque prediction.

***Index Terms*—fuzzy doctrinal concepts, legal AI, large language models, legal measurement, construct validity**

I. Introduction

Legal analysis frequently depends on concepts that are doctrinally important but empirically difficult to measure. Many such concepts are texturally open, fact-sensitive, and normatively contested: they do not map cleanly onto a fixed label, and their boundaries are often precisely what is at stake in legal argument. In these settings, the central research problem is not simply classification. It is measurement. Researchers may be able to describe the concept in doctrinal prose, yet still lack a defensible way to translate it into a structured empirical object that can be compared across cases, audited by others, and evaluated against external evidence. This problem becomes especially acute when the target concept concerns legal boundary drift: the issue is not merely whether a case is lawful or unlawful, but how strongly it exhibits features that push doctrine toward expansive or unstable application. In the language of measurement methodology, the challenge is one of valid operationalization: whether observations and scores meaningfully capture the concept they claim to measure [1].

For contested legal constructs, the difficulty often runs deeper than sparse annotation. A single stable gold standard may be unavailable not only in practice but also in principle, because gray-zone cases may admit persistent and reasonable disagreement. Treating one downstream judgment source as definitive can therefore distort the very object one aims to study. What must be validated is not merely whether a system reproduces a label, but whether its observable components and resulting scores meaningfully capture the legal concern they claim to represent. In this sense, disagreement is not just noise around the object of measurement; in part, it is constitutive of the object itself.

Recent advances in large language models make this problem newly tractable, but they do not solve it automatically. Large language models can extract structured information from legal texts at scale, and recent legal NLP benchmarks show that they can perform nontrivial legal-language tasks with impressive breadth [2], [3]. Yet model capability does not by itself determine what should count as a legally meaningful indicator, how multiple indicators should be combined, or how the resulting score should be assessed when no uncontested benchmark exists. If the entire task is delegated to an opaque end-stage predictor, the researcher may obtain an output without obtaining a defensible measure. In high-stakes legal settings, that tradeoff is difficult to justify [4].

This paper addresses that methodological problem. It develops and validates a theory-driven, auditable framework for the continuous measurement of contested legal constructs under weak-gold-standard conditions. The framework treats fine-grained semantic indicators as the primary measurement units, organizes them into explicit legal dimensions, and aggregates them through preset rules into a continuous index. As a stress-test case, the paper applies this framework to expansion risk in the Chinese offense commonly translated as "picking quarrels and provoking trouble," a doctrine long associated with concerns about elasticity, boundary ambiguity, and over-expansive application. A stylized example helps anchor the issue: a street-side altercation growing out of a specific debt or neighborhood dispute may still be redescribed in public-order language and punished under this offense rather than treated as an ordinary private conflict. The case study is not intended

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to suggest that the framework is uniquely tied to a single offense. It is used instead to show, under demanding doctrinal conditions, how a fuzzy legal concept can be transformed into a measurable object while retaining interpretability.

This paper makes three contributions. First, it develops a theory-driven and auditable framework for measuring contested legal constructs, translating a fuzzy doctrinal concern into explicit legal dimensions and a continuous, interpretable index rather than a hard binary label. Second, it proposes a layered validation design for weak-gold-standard legal settings, showing how blind human audit, prompt-based direct AI comparison, corpus-scale deployment, and robustness checks can together provide convergent evidence for construct credibility. Third, using expansion risk in Chinese "picking quarrels and provoking trouble" adjudication as a stress-test case, it demonstrates that the framework can recover human-relevant risk structure, scale to large judicial corpora, and register historically meaningful variation in legal context.

The paper does not seek to identify a final ground truth for a disputed doctrinal object, nor to offer a causal account of doctrinal change. Its contribution is narrower but methodologically important: it shows that contested legal constructs can nevertheless be operationalized into auditable and empirically testable continuous measures under weak-gold-standard conditions.

II. Background and Related Work

Contested doctrinal constructs create a measurement problem because they typically compress several layers of legal judgment at once. Courts may rely simultaneously on statutory language, assumptions about statutory purpose, tacit views about the public or private character of a dispute, and practical assessments of procedural posture or social harm. The construct of interest is therefore rarely observed directly. It is inferred from a bundle of textual and factual signals that may point in different directions. Traditional coding strategies often collapse this complexity into a single binary outcome, but doing so can erase precisely the gray-zone cases that matter most when the research question concerns doctrinal elasticity or boundary drift. In this kind of setting, measurement validity cannot be reduced to predictive convenience alone [1].

The relevant methodological question is therefore not simply whether a concept can be classified, but whether it can be represented in a graded and auditable form that preserves variation in boundary pressure. A continuous measure is useful because it can distinguish among clearly ordinary cases, clearly troubling cases, and intermediate cases in which legal judgment remains difficult. That distinction is especially important when disagreement is not incidental but partly constitutive of the object being measured.

This paper sits at the intersection of three lines of work. The first concerns concept formation and measurement validity [1]. The second concerns legal NLP benchmarks, which reveal both the promise and unevenness of model performance in law [2], [3]. The third concerns the broader role of interpretable structures in legal AI and legal analytics [5]–[9]. The present paper differs from end-task benchmark papers in two respects. It treats large language models primarily as extraction infrastructure rather than as final decision makers, and it asks how a measurable legal object should be built when the target is a contested doctrinal construct rather than an already operationalized label.

III. Framework Construction

A. Four-Dimensional Framework

The framework is built around four dimensions: order thinness, private-dispute intensity, contestation, and overcriminalization. These dimensions are not intended as latent variables discovered by a machine-learning model. They are theory-guided analytical categories designed to capture distinct pathways through which a legal provision may drift toward expansive application. The framework therefore begins from legal reasoning and only then asks how large language models can help operationalize that reasoning at scale.

Order thinness captures whether the case record contains concrete and specific facts showing disturbance to public order, as opposed to relying primarily on abstract or conclusory public-order language. Private-dispute intensity captures the extent to which the triggering conflict appears rooted in a specific prior dispute, targeted interpersonal conflict, or a setting that is more plausibly private than genuinely public. Contestation captures whether the record contains qualification disputes, nontrivial defense challenges, or weak judicial responses to classification objections. Overcriminalization captures whether the level of violence, injury, and concrete harm appears relatively limited compared with the penal framing ultimately adopted in the judgment.

Each dimension is operationalized through a set of fine-grained semantic indicators extracted from the text. These indicators include discrete fields such as victim relationship type, location type, qualification dispute, defense strategy, public-order fact specificity, harm severity, and violence-escalation pattern. In other words, the semantic indicators are the true measurement primitives of the framework. The dimensions provide an interpretable structure for combining those primitives, and the final index is only a downstream aggregation of their weighted contributions.

The implemented scoring logic is rule-structured rather than merely thematic. Each dimension is generated from a small set of upstream fields through explicit if-then rules. Order thinness is driven primarily by the tension between public-order conclusions and concrete public-order facts, but in implementation it also absorbs contextual conditions such as scene type and target specificity, because those conditions help determine whether public-order language is doing substantive work or merely redescribing a more private conflict. Private-dispute intensity is driven by relationship type and target specificity, with an additional cumulative increment when both a prior dispute and a highly specific target are present. Contestation is driven by qualification dispute, judicial response to the dispute, and defense strategy. Overcriminalization is driven by harm severity, extreme violence, violence-escalation pattern, and property-damage pathways. Not every extracted field enters the primary score directly: some upstream fields are used

chiefly for audit interpretation, pathway summaries, and the yearly corroboration analysis rather than for the weighted four-dimensional index itself.

B. Building the Expansion Index

Once the semantic indicators are extracted, they are mapped into the four dimensions and aggregated into a continuous expansion index. The aggregation rule is preset rather than learned from the audit sample. This is an intentional design choice. In a domain defined by doctrinal ambiguity, the goal is not to train an opaque model to mimic an outcome label. The goal is to make the coding logic inspectable, revisable, and contestable.

The four dimensions are assigned unequal weights. Order thinness receives the largest share because the distinction between concrete disturbance and abstract public-order rhetoric is central to the doctrinal concern under study. Private-dispute intensity receives the second-largest share because specific prior conflict and targeted interpersonal settings often mark cases in which the offense may be doing work that overlaps with narrower legal categories. Contestation and overcriminalization receive smaller but still meaningful weights because qualification disputes, weak judicial engagement, light harms, and weakly public conduct all contribute to the overall picture of expansion risk even when they are not independently decisive.

At the raw-score stage, the framework permits both positive increments and limited penalty-side deductions in two dimensions. For example, clearly random targets can reduce order thinness, while severe harm, extreme violence, and stronger violence-escalation patterns can reduce overcriminalization. Those raw dimension scores are then lower-bounded at zero, upper-bounded at the dimension-specific maximum, and normalized before weighting. The final index itself is reported on a 0–100 scale.

In the implemented case study, the framework extracts sixteen upstream semantic indicators from each judgment and then derives dimension scores through explicit if-then rules. The final expansion index is computed on a 0–100 scale as

$$\text{Index} = 100 \times (0.45\,O + 0.30\,P + 0.10\,C + 0.15\,V), \quad (1)$$

where each component is first lower-bounded at zero, upper-bounded at its dimension-specific cap, and then normalized. This produces the paper's primary object: a continuous expansion index. For practical use, the framework can also impose explicit review bands, but those cut-points are not presented as naturally discovered boundaries inside the data. They are merely a transparent operational layer placed on top of the continuous measure.

IV. Data and Audit Design

A. Case Corpus

The case-study corpus contains 9000 first-instance criminal judgments concerning the offense of "picking quarrels and provoking trouble," balanced at 1000 cases per year from 2013 through 2021. The corpus is used as a deployment-scale stress test rather than as the primary validity benchmark. The central methodological question is whether the proposed measurement design behaves coherently once applied at scale, not whether mass deployment alone can validate the measure without an external audit layer.

For each judgment, the LLM extraction pipeline produces a structured record containing the upstream semantic indicators, short evidence quotes for each field, a confidence flag, the four dimension scores, the final expansion index, and a dominant-path summary. The extraction step therefore creates an auditable intermediate representation rather than a one-step black-box score.

B. Blind Audit Sample

The external audit layer is built on a 100-case sample drawn from the full corpus through quantile stratification on the expansion index. The sample contains five equal strata of 20 cases each: $q00$–50, $q50$–75, $q75$–90, $q90$–95, and $q95$–100. This design ensures that the audit does not over-focus on either the low-risk bulk or the upper tail alone.

Human review was conducted on a judgment-oriented packet designed for final classification rather than feature coding. Human auditors assigned one of three labels: 0 for ordinary application with no clear expansion concern, 1 for a clear expansion-risk or wrongful-criminalization concern, and 2 for a gray-zone case. In the final human audit, the distribution was 19 cases labeled 0, 43 labeled 1, and 38 labeled 2. This distribution is substantively useful because it confirms that the audit task did not collapse into a near-binary consensus; a large gray zone remained visible.

Separate from the extraction LLM used to populate the framework fields, the paper also constructs a prompt-based direct-judgment comparison. This comparison uses a stripped-down numbered packet containing the judgment text but not the framework outputs, and asks four models available through a single API interface—Doubao Seed 2.0 Pro, DeepSeek V3.2, GLM-4.7, and Kimi K2.5—to issue end-stage case judgments directly. The comparison is implemented in two task organizations. In the *single-case* condition, each case is judged independently. In the *grouped-20* condition, the same 100 audit cases are partitioned into five groups of twenty, and the model is asked to label all twenty cases in a single response. The grouped condition is included because the target labels contain an explicit gray-zone category, and such intermediate labels may depend on within-set calibration rather than on purely stand-alone case evaluation.

These direct-judgment outputs are not part of the measurement pipeline itself. They serve as a collapsed-pipeline comparison: what remains if one removes the structured semantic layer and asks models to jump straight to bottom-line doctrinal assessment. More importantly, the two prompt organizations allow the paper to probe a methodological issue that the main framework is designed to reduce: whether direct end-stage legal judgment behaves like a stable case-level measure, or whether it shifts substantially when cases are embedded in a larger reference set.

TABLE I
FRAMEWORK STRUCTURE, RULE LOGIC, UPPER CAPS, AND AGGREGATION RULE

Dimension	Main fields	Typical positive logic	Penalty-side deduction	Cap	Weight
Order thinness	public-order conclusion specific public-order facts location type victim relationship target specificity	public-order conclusion with weak concrete order facts private or semi-private scene highly specific target under thin public-order narration	clearly random or nonspecific target can reduce score	4	0.45
Private-dispute intensity	victim relationship type target specificity	specific prior dispute known or targeted victim cumulative increment when dispute and target specificity co-occur	none	5	0.30
Contestation	qualification dispute judicial response defense strategy	explicit qualification dispute weak judicial response to dispute no-crime or reclassification defense	none	4	0.10
Overcriminalization	harm severity property-damage pathway extreme violence violence-escalation pattern	relatively light harms property-loss pathway with limited bodily harm	severe harm extreme violence stronger escalation	4	0.15

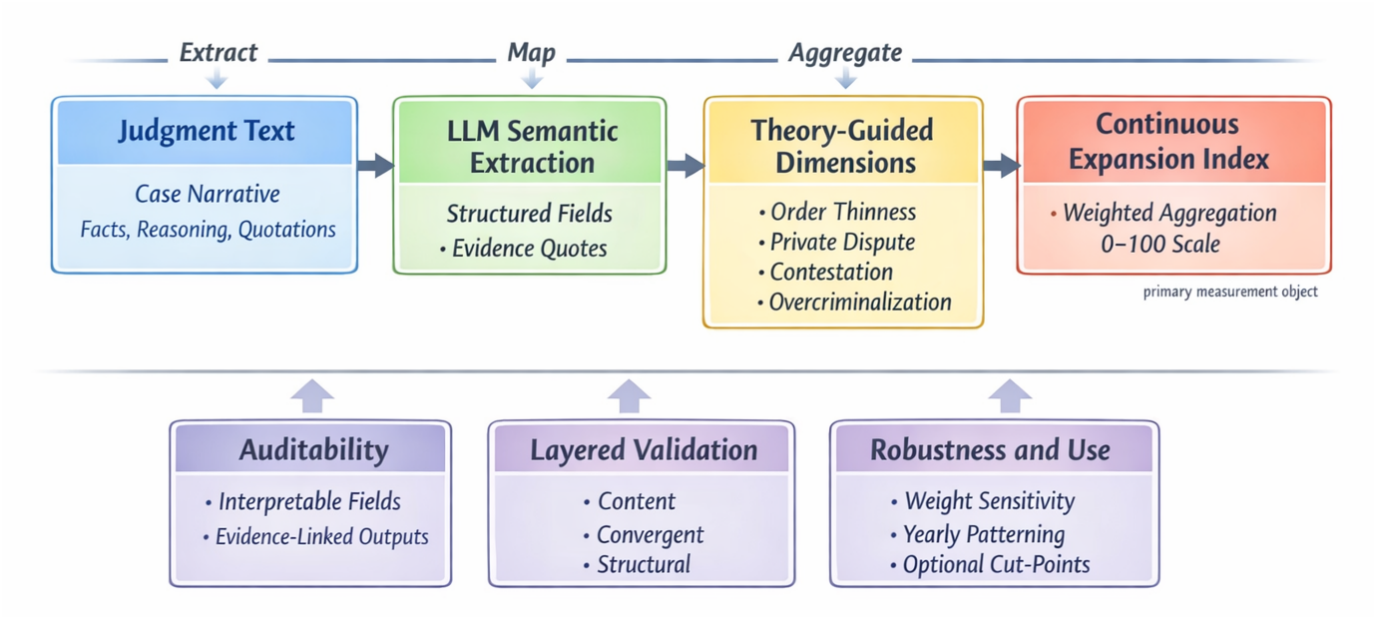

Fig. 1. Framework pipeline from judgment text to auditable semantic fields, theory-guided dimensions, a continuous index, and layered validation.

C. Blindness and Validation Logic

The blind-audit layer is designed as convergent external evidence rather than as internal rubric reproduction. Human auditors were asked to make end-stage judgments about the cases, while the framework itself was built from upstream semantic indicators and preset aggregation rules. This separation is methodologically important. If the auditors were simply asked to reenact the same scoring rubric, the exercise would show reproducibility but not external validity.

By contrast, the present design asks whether an independently produced measure orders cases in a way that aligns, at least in part, with external downstream human judgment. The point is not to install those judgments as a universal gold standard, nor to claim that they directly validate each internal field or dimension. It is to test whether the framework's output behaves coherently against assessments generated outside the scoring pipeline itself. The direct-judgment AI baseline plays a different role: it is a comparison condition showing what is lost when one collapses the framework and asks models for end-stage answers without structured semantic decomposition.

V. EVALUATION STRATEGY

The evaluation strategy follows the logic of the measurement problem. Because the target object is a contested doctrinal construct rather than a clean benchmark label, the paper uses a layered evidentiary strategy instead of a single accuracy-against-gold-standard test. The purpose of the results section is therefore not to produce one decisive score, but to ask whether different kinds of evidence point in the same methodological direction.

The first layer is content validity. This resides in the explicit field design, the four dimensions, the raw-score rules, the caps, and the aggregation logic. In other words, the framework exposes the bridge between concept and observation instead of hiding it inside a learned predictor. The second layer is convergent external validity. Here the question is not whether

auditors ratify every internal component, but whether the finished index has meaningful ordering behavior against external human judgments generated outside the scoring pipeline.

The third layer is comparative validity. This is where the paper introduces a collapsed-pipeline comparison—direct end-stage AI judgment without semantic decomposition—to clarify what is gained by structured measurement. The comparison is deliberately split into two prompt organizations, single-case and grouped-20, because a three-level legal judgment with an explicit gray zone may depend on whether cases are evaluated in isolation or against a local reference set. The fourth layer is structural and deployment validity. The 9000-case corpus is evaluated through distributional diagnostics: the overall score distribution, yearly movement in the continuous index, the concentration of the upper tail, and the dominant expansion pathways among higher-score slices. The fifth layer is robustness, including alternative weighting schemes, dimension-drop ablations, and optional cut-point checks. These layers speak to different aspects of framework credibility rather than all answering the same question. Read together, they do not claim to settle the doctrine; they test whether the framework behaves like a credible measure.

VI. Results

A. Convergent External Validation: Human Blind Audit

The results begin with the strongest external convergence test: the human blind audit. The audit does not directly validate each internal semantic field or each dimension-level score. What it tests is whether the finished index behaves like a legally meaningful measure at the output level rather than merely projecting its own internal rubric. Under the merged-12 screening view, the index reaches an AUC of 0.732 and a Spearman correlation of 0.316 with the human audit ($p = 0.0013$). Under the stricter drop-2 view, in which only clear ordinary cases and clear expansion-risk cases are retained, the signal becomes stronger: AUC rises to 0.753 and Spearman to 0.406 ($p = 0.0011$). On the full three-level scale, the human audit still shows a positive monotonic association with the index ($\rho = 0.334$), with macro AUC values of 0.559 (one-vs-rest) and 0.585 (one-vs-one).

These results matter for two reasons. First, they show that the index is not merely sorting cases according to its own internal logic; it is also tracking a nontrivial share of external human judgment generated outside the scoring design. Second, the stronger performance in the drop-2 view suggests that the framework is especially useful for ordering clearer cases, while still preserving a gray zone that the human auditors themselves did not collapse into a binary answer.

Table II also makes clear why these rank correlations should be read as conservative rather than ceiling-level summaries of alignment. The human audit is a deliberately coarse instrument: it compresses the sample into only three categories, and in the merged-12 view into only two. The framework output, by contrast, is continuous. This asymmetry creates a large tie burden on the human side before the index is even compared against it. Under the merged-12 view, 68.9% of all case pairs are tied in the human labels, while only 4.9% are tied in the index; only 29.9% of all pairs remain fully comparable after ties are taken into account. Even on the full three-level scale, 35.9% of all pairs are tied in the human labels. Kendall's τ_b is therefore especially informative here because it records how much ordering signal survives once that compression is made explicit. In substantive terms, the table shows that the framework is being evaluated against a low-resolution external ordering, so the reported rank association should be understood not as the ceiling of possible alignment, but as the amount of structure that remains visible after human-side coarsening.

Figure 2 then shows what that surviving structure looks like in score space. The continuous index is not simply loosely correlated with the human categories in the abstract; it produces a visibly ordered stratification. Cases labeled as ordinary applications cluster in the lower-score region, gray-zone cases occupy an intermediate band, and clear expansion-risk cases shift higher still. The pattern is not one of clean class separation. A fuzzy legal construct should display overlap together with ordered displacement: the ordinary category should sit lower, the gray zone should rise upward without detaching completely, and the clearest expansion-risk cases should shift higher again.

The distributional shape makes the same point in more concrete terms. The lower category forms a denser low-score body, while the gray-zone and expansion-risk categories broaden and lift upward into the middle and upper portions of the scale. The median for the ordinary category is 21.0, compared with 44.65 for gray-zone cases and 49.0 for clear expansion-risk cases. Pairwise rank-based comparisons point in the same direction. The ordinary category differs significantly from both the gray-zone category ($p = 0.0109$) and the clear expansion-risk category ($p = 0.0016$), while the gray-zone and expansion-risk categories remain only partially separated ($p = 0.0957$).

B. Collapsed-Pipeline Comparison: Direct End-Stage AI Judgment

The prompt-based direct-judgment comparison produced a more revealing result than a simple baseline table. In the controlled *single-case* condition, each model was asked to judge one case at a time under a uniform API setting. Under that organization, the three-level task collapsed strongly toward binary behavior. DeepSeek V3.2 produced 84 labels of 0 and 16 labels of 1 with no gray-zone cases at all; Doubao Seed 2.0 Pro produced 91 labels of 0 and 9 labels of 1, again with no gray-zone cases; even GLM-4.7 and Kimi K2.5 produced only 5 and 4 gray-zone labels respectively. Under this isolated-call condition, all four models remained weak relative to the framework. Their agreement with the human blind audit stayed low (0.24–0.28), and their correlations with the framework index remained small or statistically insignificant. For example, single-case DeepSeek reached only $\rho = 0.133$ against the index under the merged-12 screening view, while single-case Doubao reached $\rho = 0.064$.

The grouped-20 condition changed that picture. Once the same models judged twenty cases jointly, the gray-zone category reappeared across all four models: DeepSeek V3.2

TABLE II
HUMAN BLIND-AUDIT VALIDATION AGAINST THE EXPANSION INDEX

Human-audit view	Spearman ρ	Kendall τ_b	Human ties	Index ties	Comparable pairs	AUC	Mann-Whitney p
Merged-12 (0 vs. 1/2)	0.316	0.266	68.9%	4.9%	29.9%	0.732	0.0017
Drop-2 (0 vs. 1 only)	0.406	0.341	56.8%	4.7%	41.9%	0.753	0.0016
Full three-level scale	0.334	0.276	35.9%	4.9%	61.2%	–	–

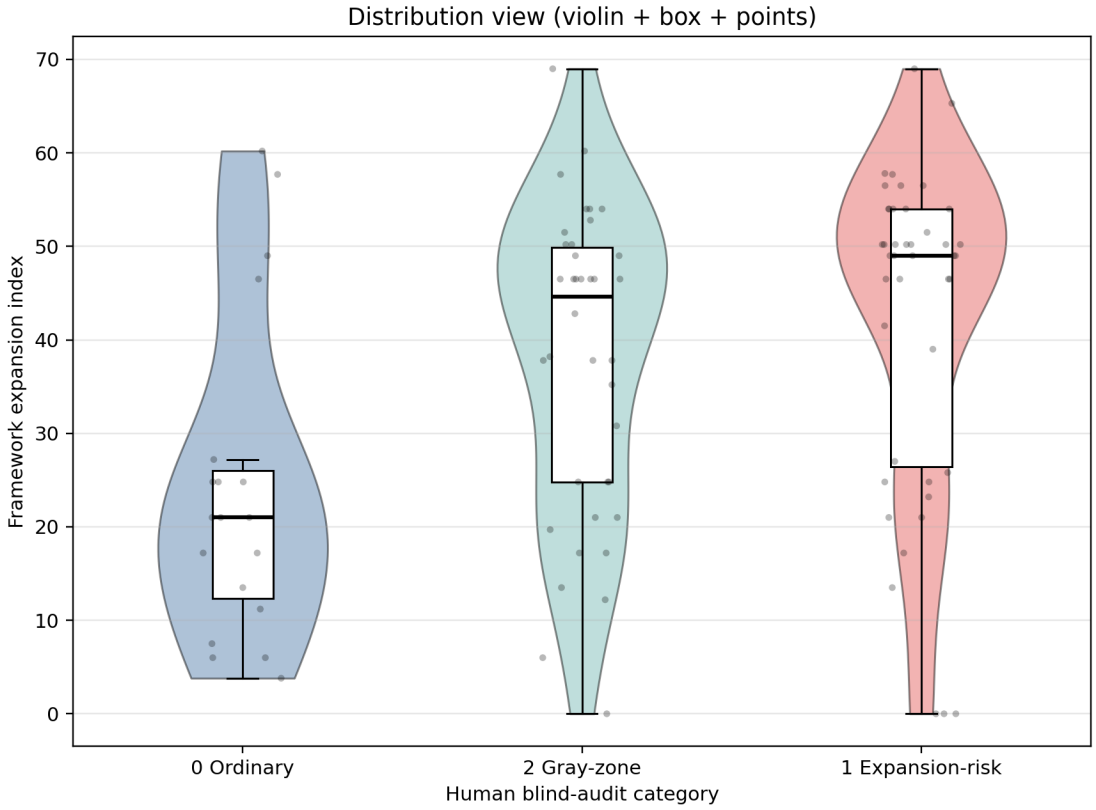

Fig. 2. Framework index stratified by the three human blind-audit categories. The figure combines a violin plot, a box plot, and case-level points to show both ordered displacement and substantial overlap: ordinary cases cluster lower, gray-zone cases occupy an intermediate band, and clear expansion-risk cases shift higher while remaining partially entangled with the gray zone.

yielded 15 gray-zone labels, Doubao Seed 2.0 Pro 17, GLM-4.7 28, and Kimi K2.5 23. This is not a trivial formatting difference. It suggests that direct end-stage judgment on a fuzzy legal construct may depend heavily on within-set comparison rather than on a stable stand-alone case metric. In other words, the models appear more capable of relative calibration across a case set than of producing a task-invariant absolute position for each single case.

That grouped behavior also changed the comparative results. Against the human audit, label-level agreement remained modest, but ordering behavior improved once gray-zone labels were treated as a middle band rather than as a third ordinal endpoint above the positive class. Under this more semantically appropriate mapping ($0 \mapsto 0$, $1 \mapsto 1$, $2 \mapsto 0.5$), grouped Doubao reached a human-alignment Spearman correlation of 0.315 ($p = 0.0014$), while grouped DeepSeek and grouped Kimi reached 0.245 ($p = 0.014$) and 0.250 ($p = 0.012$), respectively. Against the framework index, grouped DeepSeek and grouped Kimi also became more informative than their single-case counterparts. Under the merged-12 screening view, grouped DeepSeek reached $\rho = 0.210$ with AUC $= 0.636$ ($p = 0.036$), while grouped Kimi reached $\rho = 0.214$ with AUC $= 0.630$ ($p = 0.033$). Under the stricter drop-2 view, grouped Kimi reached $\rho = 0.225$ with AUC $= 0.685$ ($p = 0.049$).

Direct end-stage AI judgment is not merely weaker or stronger than the framework in a single fixed sense. It is also markedly more sensitive to how the task is organized. The same model family can look nearly binary under isolated case calls and substantially more graded under grouped judgment. That task-format sensitivity is informative in itself. It suggests that collapsed-pipeline model judgment is often acting as a context-dependent relative ranker.

TABLE III
PROMPT-BASED DIRECT AI JUDGMENT UNDER SINGLE-CASE AND GROUPED-20 CONDITIONS. HERE ρ_{MID} DENOTES SPEARMAN CORRELATION UNDER THE SEMANTICALLY ORDERED MAPPING $0 \mapsto 0$, $2 \mapsto 0.5$, AND $1 \mapsto 1$, TREATING THE GRAY-ZONE LABEL AS AN INTERMEDIATE BAND RATHER THAN AS A THIRD ENDPOINT ABOVE THE POSITIVE CLASS.

Model	Predicted label counts (0 / 1 / 2)	Human ρ_{mid}	p	Index ρ (merged-12)	p	AUC (merged-12)
Panel A: single-case prompting						
Doubao Seed 2.0 Pro	91 / 9 / 0	0.139	0.167	0.064	0.529	0.564
DeepSeek V3.2	84 / 16 / 0	0.106	0.296	0.133	0.188	0.604
GLM-4.7	71 / 24 / 5	0.101	0.319	-0.051	0.615	0.468
Kimi K2.5	74 / 22 / 4	0.113	0.264	0.089	0.376	0.559
Panel B: grouped-20 prompting						
Doubao Seed 2.0 Pro	74 / 9 / 17	0.315	0.001	0.084	0.407	0.555
DeepSeek V3.2	73 / 12 / 15	0.245	0.014	0.210	0.036	0.636
GLM-4.7	49 / 23 / 28	0.157	0.118	0.070	0.488	0.540
Kimi K2.5	66 / 11 / 23	0.250	0.012	0.214	0.033	0.630

C. Deployment-Scale Structure and Historical Responsiveness

The full-corpus deployment results are informative for two distinct reasons. First, they show whether the framework remains selective once moved from the audit sample to a large judicial corpus. Second, they show whether the framework responds to historically meaningful variation in the surrounding corpus rather than producing a stationary distribution that is indifferent to changes in legal context. On the 9000-case corpus, the mean expansion index is 23.35 and the median is 21.00, with the 75th percentile at 34.5, the 90th percentile at 49.0, and the 95th percentile at 54.0. This percentile structure matters more than any single cut-point choice: it shows that the framework produces a broad continuous distribution with a relatively narrow upper tail rather than a bloated mass of uniformly high scores.

The shape of the full distribution is itself informative. Figure 3 shows that the score is not smoothly bell-shaped; it is somewhat jagged and multi-peaked because the index is rule-structured and built from coarse semantic increments. That jaggedness is not a defect of plotting. It is a visible consequence of the framework's design: cases accumulate around recurring combinations of field values. Even so, the distribution is far from degenerate. It spans a wide range, centers in the low twenties, and then thins out sharply in the upper tail, separating a large ordinary bulk from a smaller set of increasingly troubling cases without forcing a binary break.

The same point appears in the upper tail itself. The distance between the median, the 90th percentile, and the 95th percentile shows that the high-risk slice is both real and relatively concentrated. More importantly, that slice is not compositionally random. The dominant pathway across the full corpus is order thinness (46.4%), followed by private-plus-order pathways (21.0%), private pathways (11.5%), and over-criminalization pathways (11.0%). As one moves upward in the score distribution, the private-plus-order pathway becomes especially salient. Among the top 5% of the corpus, the mean raw private-dispute score rises to 4.0 out of a raw maximum of 5, and the order-thinness score also rises substantially. The upper tail is therefore structured by recurrent doctrinal pathways rather than by an undifferentiated accumulation of high scores.

The yearly evidence is best read as focused corroboration evidence rather than as a full historical dashboard. We therefore do not plot every available yearly signal. Instead, Figure 5 isolates a small set of high-sensitivity features that should be especially informative if the framework is extracting legally meaningful information rather than annual text noise. The upper panel shows a coordinated movement in the late 2010s: the overall expansion index rises, private-dispute intensity increases, contestation becomes more common, and the private-plus-order pathway gains relative prominence. Read together, these lines suggest not merely that scores drift upward, but that the higher-risk portion of the corpus becomes more closely associated with privately rooted disputes, qualification contestation, and mixed private-order framing.

The lower panel serves a different corroborative purpose. Plea-related status and prosecutor sentence-suggestion signals rise sharply after 2019 and do so in parallel. We do not treat this pattern as causal identification, but it is difficult to read as arbitrary extraction behavior. The timing is directionally consistent with two external institutional developments: the launch of the nationwide anti-crime campaign in 2018 [10] and the formal codification and subsequent rollout of the plea leniency regime from late 2018 into 2019 [11], [12]. In that sense, the figure provides side evidence that the framework's upstream extraction layer is responsive to policy-salient changes in judicial processing rather than merely reproducing a generic annual style drift.

One further caution helps interpret the yearly curves correctly. China Judgments Online was only launched nationally in mid-2013 [13], so the 2013 level is best treated as a publication-regime starting point rather than as a stable baseline comparable in all respects to later years. The substantive point of Figure 5 therefore lies less in the isolated 2013 high point than in the coordinated reconfiguration visible around 2019–2021. That reconfiguration is methodologically useful because it is selective rather than indiscriminate: a cluster of substantively risky and procedurally policy-sensitive signals moves together, which is more consistent with a framework extracting legally relevant structure than with one merely amplifying whatever textual features happen to vary over time.

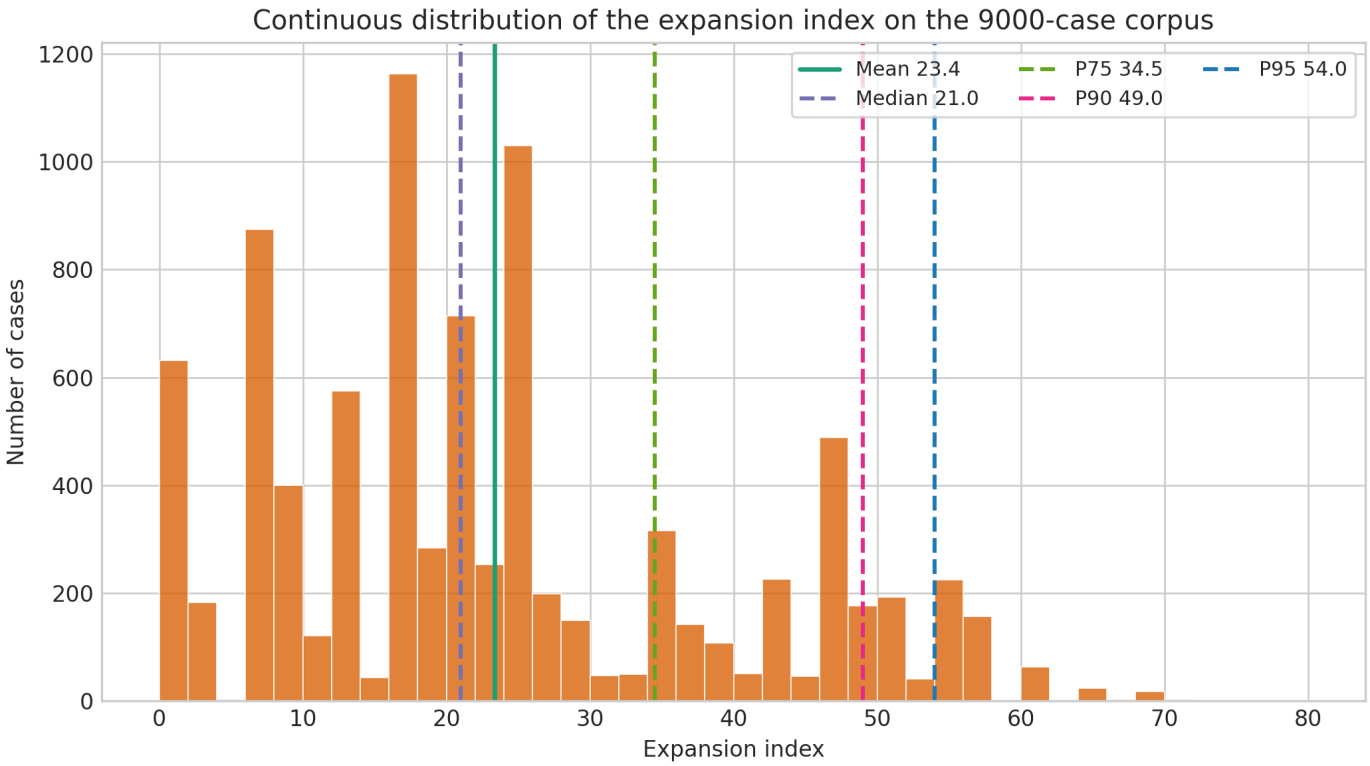

Fig. 3. Continuous distribution of the expansion index on the 9000-case corpus. The measure remains broad enough to preserve gradation while still generating a comparatively narrow upper tail.

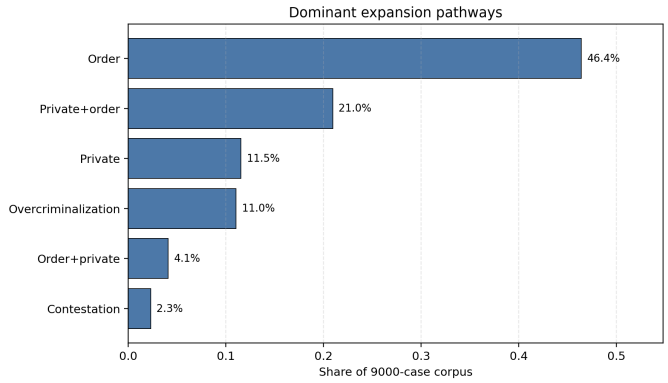

Fig. 4. Dominant expansion pathways on the 9000-case corpus.

D. Robustness

Because the index is rule-structured, it is important to ask whether the reported results depend entirely on one exact weight vector. A first sensitivity analysis therefore recomputed the index under several plausible alternatives, including equal weights, a more order-heavy version, a more private-heavy version, and a more balanced legal-design version. The main human-alignment result remains stable across these alternatives. On the 100-case human audit, Spearman correlation under the merged-12 view ranges from 0.308 to 0.318 and AUC from 0.725 to 0.734. On the 9000-case corpus, the alternative indices remain highly correlated with the baseline index, with rank correlations between 0.930 and 0.998.

This robustness check does not prove that the chosen weights are uniquely correct. It does show that the framework's main signal is not a knife-edge artifact of one arbitrary formula. Supplementary cut-point checks point in the same direction, but they remain secondary to the paper's main continuous-measurement claim.

We also explored a small family of alternative aggregation rules without changing the upstream semantic fields. These supplementary experiments included equal-weight and reweighted linear variants, mild nonlinear transforms, simple gating rules, and a limited interaction-sensitive specification. The headline result was not that a radically different aggregator solved the problem. Even aggressive in-sample searches produced only modest gains over the baseline human-audit rank alignment, which suggests that the main constraint is unlikely to lie in the final weighted sum alone. More revealingly, the better-performing exploratory variants tended to increase the relative importance of private-dispute structure, and in some cases modestly benefited from an order–private interaction. Substantively, that pattern is plausible: expansion risk often appears most acute when a dispute that looks strongly private is nevertheless redescribed in public-order language. By contrast, overcriminalization was repeatedly down-weighted by the better-performing exploratory variants, consistent with the dimension-drop result reported below. We do not adopt any of these exploratory variants as the paper's main specification, because the present paper prioritizes ex ante transparency and theory-guided auditability over in-sample optimization. Their value is diagnostic instead: they indicate that the baseline index

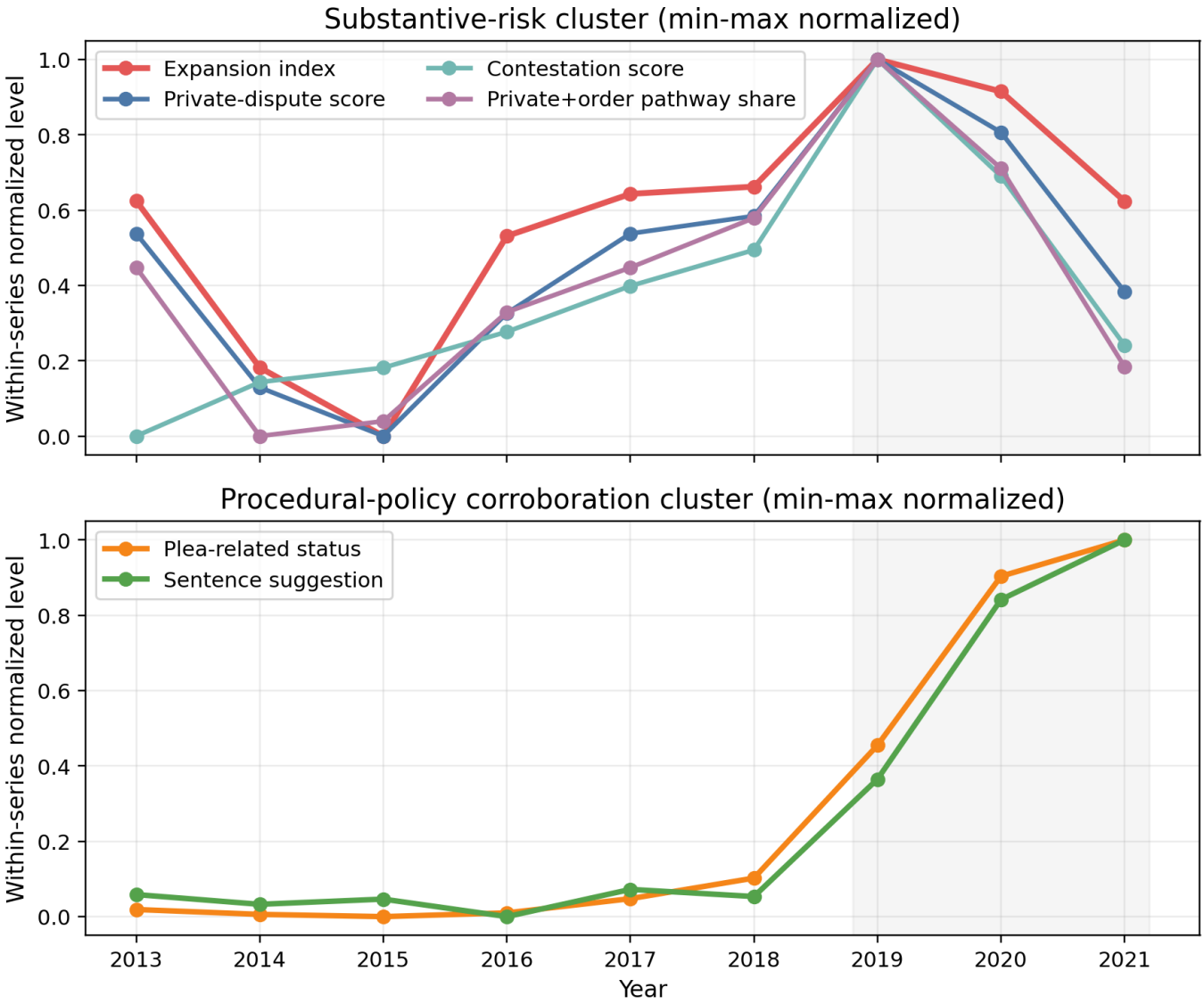

Fig. 5. Focused yearly corroboration signals on the 9000-case corpus, shown after within-series min-max normalization. The upper panel groups co-moving substantive-risk signals; the lower panel groups procedural-policy signals whose post-2019 rise is directionally consistent with the growing visibility of plea-related processing and prosecutor sentence recommendations in judgments.

is reasonably stable, while also pointing to private-dispute structure, and possibly order–private interaction, as the most promising directions for future refinement.

Another useful check is dimension ablation. Instead of merely perturbing the weights, this analysis removes one doctrinal dimension at a time and renormalizes the remaining weights. The results are informative in two ways. First, no single dimension is solely responsible for the signal. Even after dropping one dimension, the index continues to exhibit positive association with human audit. Second, the dimensions are not equally important. Removing private-dispute intensity produces the clearest deterioration in alignment, while removing contestation yields a more modest decline. Removing overcriminalization slightly improves blind-audit alignment in the present sample, which should not be read as proving that the dimension is useless. A more cautious reading is that the overcriminalization pathway may be carrying noisier or more case-specific information than the core order and private-dispute pathways, and thus deserves further recalibration in future versions of the framework.

VII. Discussion

The results support three methodological takeaways. First, a highly fuzzy doctrinal concern can be converted into a continuous measurement object without surrendering interpretability. The crucial move is not to ask a model for a mystical final score, but to force the pipeline through auditable semantic fields, explicit dimension scores, and transparent aggregation rules.

Second, for contested constructs, evaluation is better understood as a layered evidentiary strategy than as benchmark accuracy against a supposedly final label. Human blind audit supplies the strongest convergent external evidence in the

TABLE IV
DIMENSION-DROP ABLATION AGAINST THE HUMAN BLIND AUDIT

Variant	Merged-12 ρ	Merged-12 AUC	Drop-2 ρ	Drop-2 AUC	9000-case ρ vs. baseline
Baseline	0.316	0.732	0.406	0.753	1.000
Drop order thinness	0.324	0.737	0.407	0.754	0.883
Drop private dispute	0.264	0.694	0.359	0.724	0.950
Drop contestation	0.307	0.723	0.389	0.741	0.987
Drop overcriminalization	0.365	0.765	0.465	0.788	0.949

current paper, but it is not treated as a universal truth source and it does not directly validate each internal component. The direct AI blind-audit models are useful precisely as a comparison condition: they show what model performance looks like when the semantic decomposition layer is removed and the task is collapsed into end-stage judgment. The grouped-20 comparison sharpens that lesson further. Once cases are presented as a local set, gray-zone judgments become much more common and rank-order agreement with the human audit improves for several models. The key implication is not simply that grouped prompting may look better on some metrics. It is that direct model judgment appears to depend materially on reference-set calibration. For a measurement paper, that dependence matters: a tool intended to place one case on a continuous scale should not need to see a batch of neighboring cases before it can recover an interpretable middle band. On that criterion, the framework and grouped prompting are solving different problems. The framework is designed to approximate a stand-alone case-level measure, whereas grouped direct judgment behaves more like a context-dependent ranker over a local case set.

Third, the deployment evidence shows that a rule-structured index can remain selective and structurally informative at scale. The framework produces a concentrated upper tail, a patterned high-score slice, and a yearly profile that moves through interpretable internal components rather than through uninterpretable score drift. The distributional and yearly results matter together: the upper tail remains structured by recurring pathways, and the focused yearly corroboration signals move in coordinated clusters rather than as isolated annual fluctuations. The broader implication is that doctrinal fuzziness need not be treated as methodologically fatal. Many legal concepts are contested, graded, and context-sensitive. The practical challenge is therefore not to eliminate disagreement, but to represent disagreement in a measurable structure that can be debated, audited, and revised.

VIII. LIMITATIONS

The paper has several limitations, and they are closely tied to the design choices that make the framework interpretable. First, the human blind audit is best understood as convergent validation evidence, not as a universal gold standard. A 100-case audit can test whether the index shows meaningful external alignment, but it cannot settle every doctrinal disagreement, especially when the target object is itself contested. Second, the case study focuses on a particularly difficult offense in one legal system. That makes the paper a demanding methodological demonstration, but it does not by itself establish portability to every other fuzzy doctrinal concept. Third, the aggregation rule remains theory-guided and researcher-specified. The sensitivity results show that the main signal is not uniquely dependent on one exact weight vector, but they do not eliminate the need for future debate over field definitions, score caps, optional operational views, and possible rule ablations. Fourth, the yearly evidence should be read as responsiveness evidence rather than as a causal account of legal change; the framework registers historically meaningful variation, but it does not by itself identify the institutional mechanisms producing that variation. Finally, the direct end-stage AI comparison should not be over-read as a universal alternative to the framework; it is included here as a collapsed-pipeline comparison, not as a claim that direct model judgment defines the right task. The grouped-20 results make this point clearer rather than weaker: direct prompt-based judgment becomes more graded when models can compare cases within a set, which is informative in its own right but also shows that the resulting labels are sensitive to task organization. That sensitivity limits how far such outputs should be treated as stable case-level measurements.

IX. CONCLUSION

This paper develops an auditable framework for the continuous measurement of contested legal constructs with the assistance of large language models. The central move is to treat fine-grained semantic indicators as the true measurement units, to aggregate them through explicit rules into a continuous index, and then to evaluate that index through a layered evidentiary strategy rather than merely trusting the internal logic of the scoring formula or demanding a single gold-standard label.

The paper's contribution is methodological first and substantive second. It does not claim to settle the doctrinal controversy that motivates the case study. It shows instead that even a deeply fuzzy legal controversy can be converted into a structured quantitative object that is inspectable, revisable, and open to scientific evaluation under weak-gold-standard conditions. The direct AI comparison now reinforces that point in two ways. Under isolated single-case prompting, end-stage judgments collapse toward binary behavior and align only weakly with the external audit and the index. Under grouped prompting, the same model families become more graded and often more human-like in ranking behavior, but precisely because they can calibrate against a visible case set. More broadly, the paper argues that legal AI need not choose

between opaque prediction and purely qualitative doctrinal description. For a class of contested, graded, and context-sensitive legal problems, auditable continuous measurement offers a third path: one that is more explicit than black-box prediction and less dependent on reference-set calibration than grouped direct model judgment.

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Data and Code Availability

Code and research materials for this project are publicly available at https://github.com/JialingRichard/legal-measurement. The dataset used in this study was compiled from publicly circulated collections of Chinese judgment documents covering 2013–2021 that were originally drawn from materials publicly accessible through China Judgments Online and related public-document collections. The public repository includes the curated 9000-case corpus used in this study, the framework prediction outputs, and the 100-case blind-audit materials, subject to the documentation and redistribution conditions stated in the repository.

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